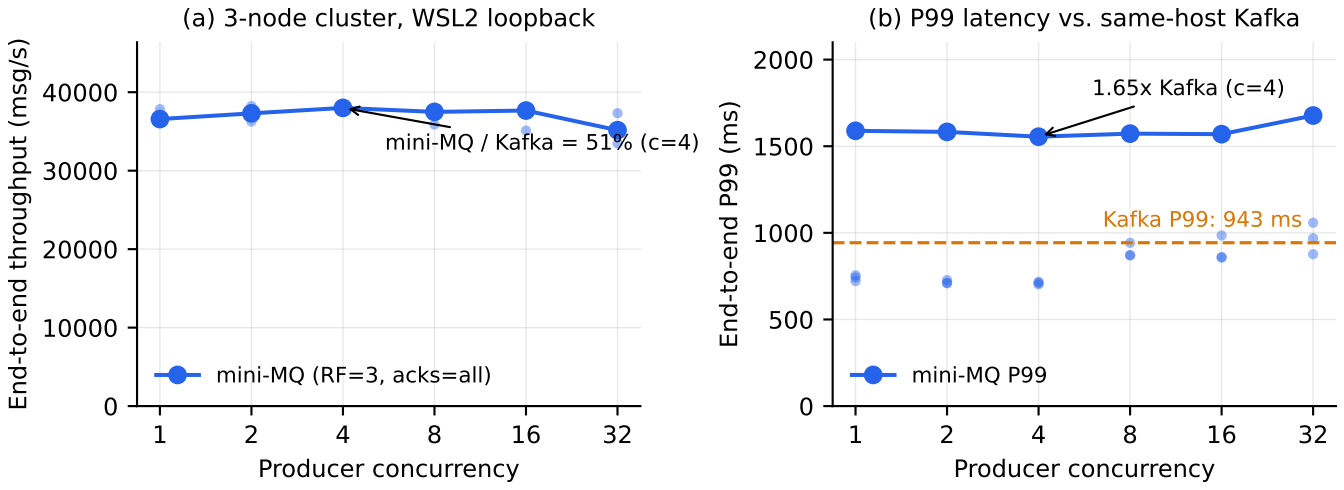
(a) 3-node cluster, WSL2 loopback
End-to-end throughput (msg/s)
40000
30000
20000
10000
0
mini-MQ / Kafka = 51% (c=4)
mini-MQ (RF=3, acks=all)
1
2
4
8
16
32
Producer concurrency
(b) P99 latency vs. same-host Kafka
End-to-end P99 (ms)
2000
1500
1000
500
0
1.65x Kafka (c=4)
Kafka P99: 943 ms
mini-MQ P99
1
2
4
8
16
32
Producer concurrency